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REVIEW

Herb-Drug Interactions: Focus on Metabolic Enzymes and Transporters

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As the uses of herbal medicines from traditional natural products are increased, the need for pharmacokinetic studies and relevant data are also increased for safe pharmacotherapy. The market entry for the traditional herbal medicine is easier compared with that for synthetic drugs because of a lower regulatory barrier. Thus, the exact mechanisms for the absorption, distribution, metabolism and excretion of active components in herbal medicines and the potential herb-drug interactions are not always fully understood. Recently, there has been an increasing interest in pharmacokinetic studies of herbal remedies and relevant data of commonly used herbal remedies are accumulating in this field. In this review, the effects of nine botanicals (ginkgo, green tea, grapes, licorice, saw palmetto, garlic milk thistle, ginseng and St. John's wort) on metabolic enzymes and transporters affecting absorption and disposition of herbal products are summarized. The source of samples (extracts and individual components), the species (human and animal) and *in vivo* and *in vitro* systems were separately reviewed for a better understanding of herb-drug interactions.

Key words: Herb-drug interaction, P450, Transporter, Pharmacokinetics

INTRODUCTION

The use of natural products as herbal medicines has increased steadily over the last decade. Herbal medicines are an important group of multi-component therapeutics and are mainly used to manage various chronic diseases and to promote health in the Western world (Stone, 2008; Patwardhan et al., 2010). To date, there are more than 11,000 species of plants for medical use and about 200 of them are commonly used (Ernst and Pittler, 2002). Moreover, approximately 80% of the population in Asian countries use herbal medicines for promoting health and managing common maladies such as cold, inflammation, pain, heart diseases, liver cirrhosis, diabetes and central nervous system disease (Ernst and Cassileth, 1998; Bent, 2008; Low Dow, 2009; Roy et al., 2010). Additionally, patients with chronic diseases, which are likely to be treated with multiple drugs, use herbal medicines more frequently, thereby increasing the risk for drug-drug interactions (White et al., 2007; Miller et al., 2008). Despite the popularity of herbal medicines, their use is largely not evidence-based because there is lack of clinical evidence for the efficacy, target and safety of most herbal medicines.

In contrast to the development of synthetic drugs, there is no regulatory need to study the pharmacokinetics and absorption, distribution, metabolism and excretion of herbal medicines (Vlietinck et al., 2009; Jordan et al., 2010; Zhou et al., 2010). Although there exists evidence for the efficacy and safety of herbal medicine, the mechanism of absorption of herbal medicines are diverse, depending on how it is tested (*in vivo* or *in vitro*), the sources of biological samples (humans or animals), the doses, the periods of administration, and so on.

In particular, metabolism of xenobiotics, including drugs and herbal medicines, has been regarded as one of the most important and complex processes in the body, leading to the excretion of most drugs and herbal medicines. Most herbal medicines undergo Phase I and/or II metabolism, yielding inactive or active metabolites. To optimize the use of herbal medicines (e.g. dosage, regimen and administration route), knowledge of their biological fate, including disposition pathways

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and kinetics in the body, is certainly needed. Pharmacokinetic studies of herbal remedies also provide important information on potential herb-herb and herb-drug interactions. However, the safety profiles of most herbal remedies in humans are unknown and different results for many herbal remedies have been reported *in vitro* and *in vivo* studies. In this regard, it is necessary to understand the pharmacokinetics, focusing on the metabolism of herbal medicines and considering *in vivo* and *in vitro* systems.

The bioactivity of herbal medicines is considered to stem from the interaction of the biological system with multiple active compounds present in herbal medicines at low levels. The exact spectrum of active compounds of most herbal medicines is not defined, although a large number of pharmacologically active compounds have been isolated and purified from various herbal medicines (Carbotto et al., 2010). In most cases, the identification of active herbal compounds is from *in vitro* studies and less from animal studies; only very limited activity data are from clinical studies. Variable amounts of active compounds in herbal preparations are a major challenge in the standardization of herbal products (Sahoo et al., 2010).

Recently, there has been an increasing interest in pharmacokinetic studies of herbal remedies and relevant data for commonly used herbal remedies have been accumulating in this field. To update our knowledge, this paper highlights the pharmacokinetic properties of herbal remedies, focusing on metabolic enzymes and transporters affecting absorption of herbal products. To retrieve relevant data, nine botanicals, widely used in Korea and ranked as top-selling products in Europe and the USA, were selected and the literature on nine botanicals with any involvement in metabolic enzymes and transporters are reviewed.

INTESTINAL ABSORPTION OF HERBAL MEDICINES

Most herbal medicines are orally administered in a chronic regimen. There are limited studies on the oral absorption and factors affecting herbal remedies in humans. Since oral absorption is a major determining process for the bioavailability of active components from plant remedies, it is necessary to understand the absorption rate, mechanism and influencing factors. It is thought that the oral absorption of herbal components resembles that of synthetic drugs, but other factors should be considered. For example, herbal medicines are a mixture of multiple ingredients with distinct physicochemical properties that may interact with each other in the gut. In this regard, it is difficult to predict the absorption rate and extent of herbal medicines. In particular, the oral bioavailability of herbal medicines are associated with many presystemic processes, including solubility in gastrointestinal fluid, membrane permeability, degradation in the gastrointestinal tract, transporter [e.g. P-glycoprotein (P-gp/MDR1/ABCB1)]-mediated intestinal efflux, presystemic gut wall metabolism and presystemic hepatic metabolism. In this regard, low or poor intestinal absorption of herbal compounds may occur, leading to a low oral bioavailability.

P-gp is expressed at high levels in the apical membrane of the intestine and thus limits the absorption of many drugs (Thiebaut et al., 1987; Dean, 2002; Zhou, 2008) and may play an important role in herbal absorption. A number of herbal compounds have been identified as substrates of P-gp.

HEPATIC METABOLISM OF HERBAL MEDICINES

Phase I reactions in liver are mainly oxidative or reductive reactions catalyzed by cytochrome P450s (CYPs), flavin-containing monooxygenases, epoxide hydrolase, carboxylesterase and amidase, peroxidase, alcohol/aldehyde dehydrogenase, monoamine oxidase, or a-nicotinamide adenine dinucleotide phosphate (NADPH) quinone reductase (Lin and Lu, 1997). Phase I reactions usually make an herbal compound more susceptible to Phase II reactions, which are conjugative reactions, such as glucuronidation catalyzed by various UDP-glucuronosyl transferases (UGTs) and sulfation by sulfotransferases (SULTs), generally producing molecules more amenable to biliary or renal excretion. As such, the disposition and clearance of herbal medicines may be altered when the enzymes that metabolize them are modulated by co-administered drugs or herbal medicines.

INTESTINAL METABOLISM OF HERBAL MEDICINES

CYP- and UGT-mediated metabolism of herbal compounds occur in the intestine and this may play an important role in determining their oral absorption and bioavailability. Multiple CYPs (in particular CYP3A4) and UGTs are present in the intestine at substantial levels and herbal compounds are subject to presystemic oxidative and/or conjugating metabolism following oral administration. Intestinal CYP3A4 can serve independently of the liver as a highly efficient metabolic barrier during the uptake of various herbal ingredients from the intestine.

EFFECTS OF INDIVIDUAL HERBAL MEDICINES ON METABOLISM AND TRANSPORTER ENZYMES

Ginkgo

Ginkgo biloba leaves have been used to ameliorate blood circulation and neurodegenerative disorders relevant to memory (Colalto, 2010). Due to its perceived pharmacological benefits, ginkgo products are being sold throughout the world as dietary supplements or OTC drugs. It was also reported that ginkgo products were ranked as one of the top-selling dietary supplements in the USA and Europe (Mohamed and Frye, 2011a). Ginkgolides and bilobalides are recognized as representative secondary metabolites of this plant.

Ginkgolides A, B, C, J, and bilobalide were found to inhibit weakly or negligibly CYP1A2, CYP3A, and CYP2C9 in human liver microsomes, while kaempferol, quercetin, apigenin, myricetin, and tamarixetin exerted inhibited CYP1A2 and CYP3A. Quercetin, amentoflavone (biflavonoid), and sesamin (lignan) were inhibitors of CYP2C9 (von Moltke et al., 2004). *G. biloba* extract EGb761 was assessed against *in vitro* human CYP1A2, 2C9, 2D6, 2E1, and 3A4 for its inhibitory activity and found to inhibit CYP1A2 and 2E1. Further fractionation of EGb761 into two subfractions, a terpenoidic fraction and a flavonoidic fraction, was conducted and these two fractions were evaluated for their effect on CYPs. The flavonoidic fraction inhibited CYP2C9, 1A2, 2E1, and 3A4 whereas the terpenoidic fraction inhibited only CYP2C9 of the CYPs used in this study (Gaudineau et al., 2004). When *G. biloba* extract was orally administered to rats, CYP1A2 activity could be induced. Therefore, this extract increased the clearance of theophylline (Tang et al., 2007). Using recombinant human CYP isoforms, bilobalide, ginkgolides A, B, C, and ginkgolic acids I, II were tested for their effect on CYP isoforms including CYP1A2, 2C9, 2C19, 2D6, and 3A4. Only ginkgolic acids I and II inhibited all CYP isoforms used in this assay (Zou et al., 2002). A standardized preparation (120 mg, bid) of *G. biloba* was given to normal volunteers (n = 12) and changes in CYP2D6 (dextromethorphan) and 3A4 (alprazolam) activity were monitored. Unlike what was mentioned above, the extract did not affect CYP2D6 or CYP3A4 pathways of elimination

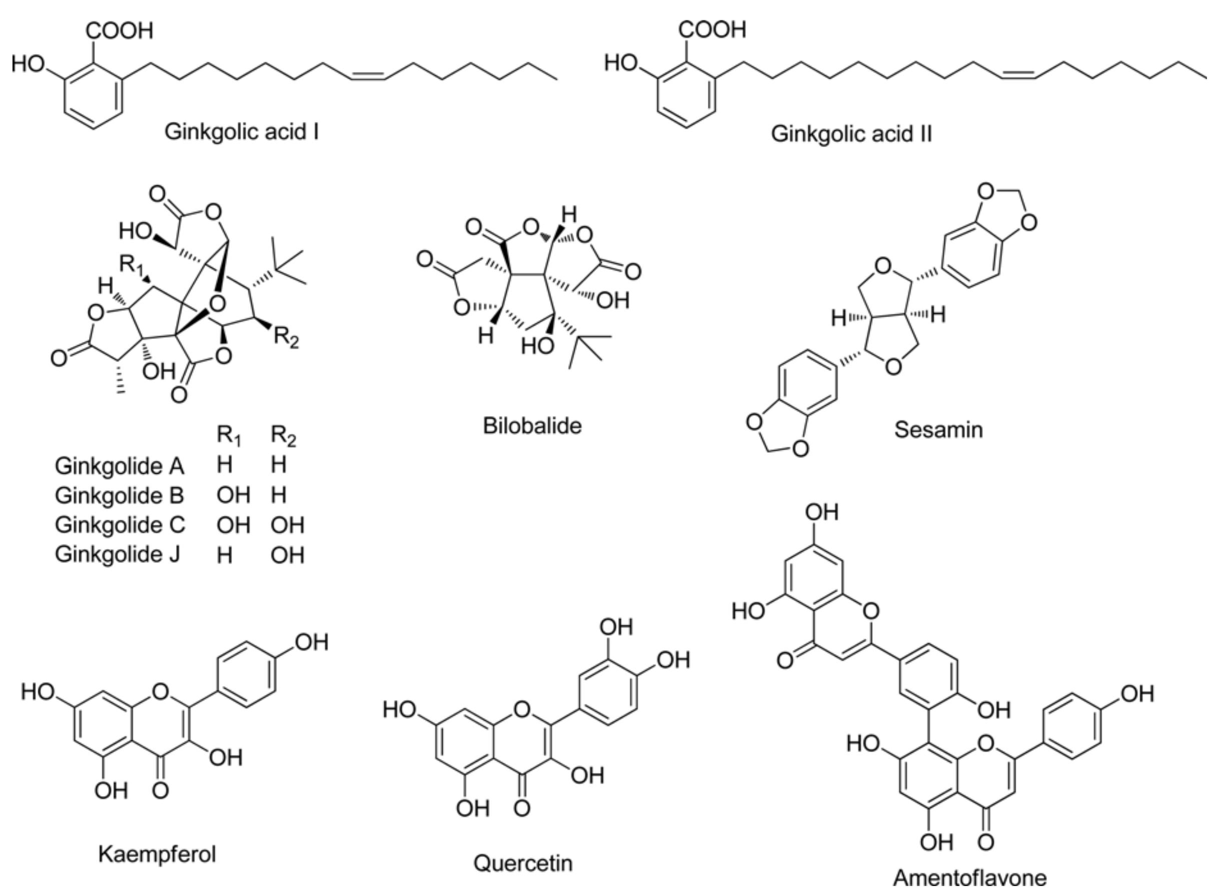

Fig. 1. The structures of the main active components of ginkgo

Table I. Effects of ginkgo extract and its active components on metabolic enzymes and transporters

Plant	*In vitro*/ *In vivo*	Metabolism	Transporter	References
Ginkgo				
Extract	*In vivo* (R)	CYP1A2 (↑)		Tang et al., 2007
	In vivo (H)	CYP 2D6 (↔) CYP3A4 (↔)		Markowitz et al., 2003a
	In vitro (H)		P-gp (↓)	Hellum and Nilsen, 2008
	In vitro (R)		OATP-B (↓)	Najar et al., 2010
Extract (EGb761)	*In vitro* (H)	CYP1A2 (↓) CYP2E1 (↓)		Gaudineau et al., 2004
Flavonoid fraction	*In vitro* (H)	CYP2C9 (↓) CYP1A2 (↓) CYP2E1(↓) CYP3A4 (↓)		Gaudineau et al., 2004
Terpenoid fraction	*In vitro* (H)	CYP2C9 (↓)		Gaudineau et al., 2004
Ginkgolides A, B, C, J, bilobalide	*In vitro* (H)	CYP1A2 (↔) CYP3A (↔) CYP2C9 (↔)		von Moltke et al., 2004
Ginkgolides A, B, bilibalide	*In vitro* (H)	Glucuronidation (↔)		Mohamed and Frye, 2010
Ginkgolic acids I, II	*In vitro* (H)	CYP1A2 (↓) CYP2C9 (↓) CYP2C19 (↓) CYP2D6 (↓) CYP3A4 (↓)		Zou et al., 2002
Kaempferol, quercetin, apigenin, myricetin, tamarixetin	*In vitro* (H)	CYP1A2 (↓) CYP3A (↓)		von Moltke et al., 2004
Quercetin, amentoflavone, sesamin	*In vitro* (H)	CYP2C9 (↓)		von Moltke et al., 2004
Quercetin, kaempferol	*In vitro* (H)	Glucuronidation (↓)		Mohamed and Frye, 2010
	In vitro (H)	UGT1A3 (s) UGT1A9 (s)		Oliveira and Watson, 2000; Chen et al., 2008
kaempferol	*In vitro* (D)		BCRP (↓), (s)	An et al., 2011

(Markowitz et al., 2003a). It was reported that ginkgo extract and its main flavonoids may modulate UGT enzymes (Mohamed and Frye, 2011a). The unhydrolyzed and acid-hydrolyzed ginkgo extracts, and their components were incubated with mycophenolic acid in human liver microsomes and human intestinal microsomes to examine the formation of mycophenolic acid-7-*O*-glucuronide. Ginkgolides A and B, and bilobalide did not affect glucuronidation in human liver and intestinal microsomes. Quercetin seemed to be a mixed-type inhibitor and kaempferol was a noncompetitive inhibitor in human liver microsomes, and a mixed-type inhibitor in human intestinal microsomes (Mohamed and Frye, 2010) The extracts inhibited glucuronidation of mycophenolic acid in human microsomes, suggesting the main flavonoids may be in part responsible for this inhibition (Mohamed and Frye, 2010). Previous studies proved that kaempferol and quercetin were substrates for UGT1A3 and UGT1A9 and their glucuronides were observed in *in vitro* tests (Oliveira and Watson, 2000; Chen et al., 2008). *G. biloba* extract was tested in Caco-2 cells to assess its inhibitory activity against P-gp using digoxin flux; it inhibited P-gp (Hellum and Nilsen, 2008). In addition, kaempferol increased the efflux of quercetin as a BCRP inhibitor and was also observed as a substrate for BCRP, indicating a possible correlation with the low bioavailability of kaempferol (An et al., 2011). In addition, OATP-B mediated uptake of estrone-3-sulfate was inhibited by *G. biloba* extract, suggesting that influx of any molecules may be hindered by the extract (Najar et al., 2010). All chemical structures mentioned in this section, and the relevant metabolic enzymes and transporters are summarized in Fig. 1 and Table I, respectively.

Green tea

The leaves of *Camellia sinensis* are being used as a popular source of beverages and dietary supplements. Moreover, its purified extract was approved as the first botanical drug by the US FDA in 2006. As beverages containing *C. sinensis* leaves are being consumed daily by the public, the extract of *C. sinensis* has been investigated for its effects on metabolic enzymes and transporters (Colalto, 2010). Purine alkaloids (caffeine, theophylline) and flavonoids (catechin, epicatechin, (–)-epigallocatechin-3-gallate (EGCG) and (–)-epicatechin-3-gallate (ECG)) are known as the

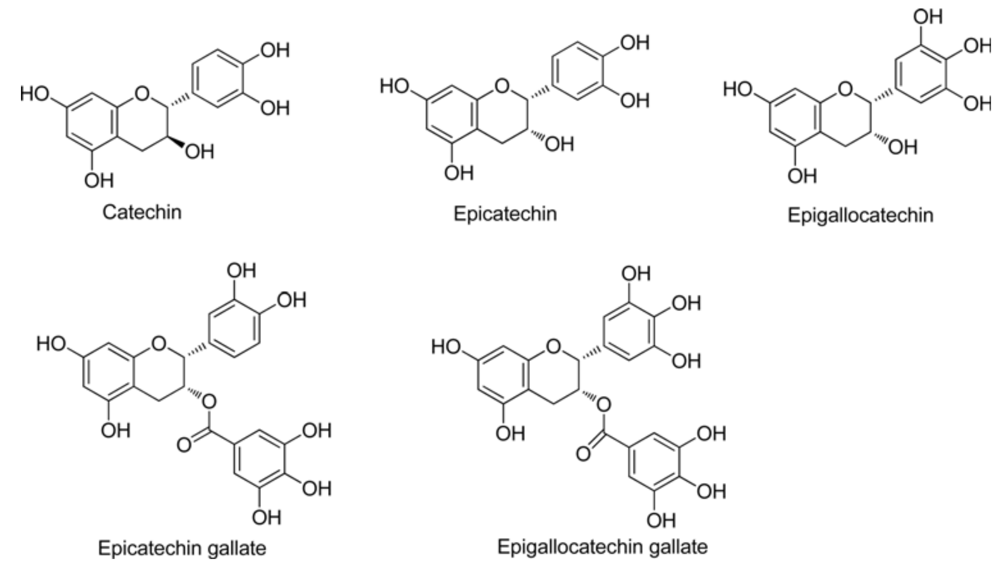

Fig. 2. The structures of the main active components of green tea

Table II. Effects of green tea extract and its active components on metabolic enzymes and transporters

Plant	*In vitro/In vivo*	Metabolism	Transporter	References
Green tea				
Extract	*In vitro* (H)	CYP2C9 (↓) CYP2D6 (↓) CYP3A4 (↓)		Nishikawa et al., 2004
	In vivo (R)	CYP3A (↑)		Nishikawa et al., 2004
	In vitro (H)	CYP1A2 (↑)		Netsch et al., 2006
	In vitro (D)		MRP2 OATP-B	Netsch et al., 2005
	In vitro (H)		OATP-B (↓)	Fuchikami et al., 2006
EGCG	*In vitro* (H)	UGT1A4 (↓)		Mohamed and Frye, 2011b
	In vitro (H, R, I)	UGT1A1, 1A8, and 1A9 (s)		Lu et al., 2003
EGCG and ECG	*In vitro* (R)			Crespy et al., 2004
	In vitro (D)		MRP2 (↔)	Netsch et al., 2005
	In vitro (H)		OATP-B (↓)	Fuchikami et al., 2006
EGCG, ECG, and CG	*In vitro* (Ham)		P-gp (↓)	Jodoin et al., 2002
Catechin	*In vitro* (R)		P-gp (↓)	Najar et al., 2010

main constituents of *C. sinensis* leaves.

Green tea extracts were evaluated for CYP2C9, 2D6, and 3A4 activities in human liver microsomes and were found to inhibit all the tested enzymes (Nishikawa et al., 2004). Moreover, this extract was given to rats to examine the effect on CYP3A (midazolam). Green tea extracts induced CYP3A in the liver, but rather reduced levels of CYP3A in the small intestine. This observation may not be predictable from *in vitro* experiments (Nishikawa et al., 2004). Also green tea extract appeared to induce CYP1A2 mRNA expression, but not CYP1A1 and 3A4 expression in LS-180 cells, a human gastrointestinal epithelial cell line (Netsch et al., 2006). Of green tea catechins, (–)-epigallocatechin-3-gallate (EGCG) and (–)-epicatechin-3-gallate (ECG) were metabolized less to their glucuronides than quercetin in rat intestinal microsomes (Crespy et al., 2004). Further testing of EGCG on UGT1A4, 1A6, and 1A9 activity showed that EGCG is a potential inhibitor of UGT1A4 (Mohamed and Frye, 2011b). Glucuronidation studies for EGCG and EGC were conducted using human, mouse, and rat microsomes as well as in nine different human UGT1A and 2B isozymes expressed in insect cells (Lu et al., 2003). EGCG was highly glucuronidated by UGT1A1, 1A8, and 1A9. Green tea polyphenols inhibited P-gp efflux activity. Of the polyphenols present in tea, EGCG, ECG, and CG participated in suppressing P-gp activity (Jodoin et al., 2002). Also, catechin inhibited P-gp ATPase activity in rat jejunal membranes (Najar et al., 2010). MRP2 activity was inhibited by green tea extract, but its components, EGCG and EGC, did not affect MRP2 activity (Netsch et al., 2005). Moreover, green tea extract inhibited OATP-B mediated uptake. Among the five catechins used in the assay, ECG and EGCG were found to lower the uptake through OATP-B (Fuchikami et al., 2006). All chemical structures mentioned in this section, and the relevant metabolic enzymes and transporters are summarized in Fig. 2 and Table II, respectively.

Grapes

Grapes (*Vitis vinifera*) are cultivated as an edible fruit and for making wine. Polyphenolic compounds including flavonoids and resveratrol are well-known components of grapes. In particular, resveratrol has been studied in a wide range of pharmacological assays and has the potential for human health improvement.

Red wine solids and resveratrol were investigated for their ability to modulate CYPs. Both samples were found to irreversibly inhibit CYP3A4, and reversibly inhibit CYP2E1 in a non-competitive manner. When compared with the inhibitory potency of red wine solids, resveratrol did not appear to be responsible for the observed inhibitory activity. Resveratrol also inactivated CYP3A4 in insect microsomes containing human CYP3A4 (Piver et al., 2001). CYP3A4 inactivation by a variety of red wines was not correlated with the content of resveratrol. (Chan and Delucchi, 2000). Clinical investigation of any change in CYPs by resveratrol from 42 healthy volunteers was carried out. CYP3A4, 2D6, and 2C9 activities were inhibited while CYP1A2 was induced (Chow et al., 2010). UGT1A1 activity was not influenced by resveratrol in this study (Chow et al., 2010). A dimer of resveratrol, ε-viniferin, was found to be involved in downregulating human CYP1A1, A2, B1, 2A6, 2B6, 3A4 and 4A activities except for CYP2E1 (Piver et al., 2003). In particular, a furanocoumarin, bergamottin, inactivated the enzymatic activity of CYP 1A2, 2A6, 2C9, 2C19, 2D6, 2E1, and 3A4 in human liver microsomes (He et al., 1998). The involvement of UGT isoforms in metabolizing resveratrol was introduced by Brill et al. (Brill et al.,

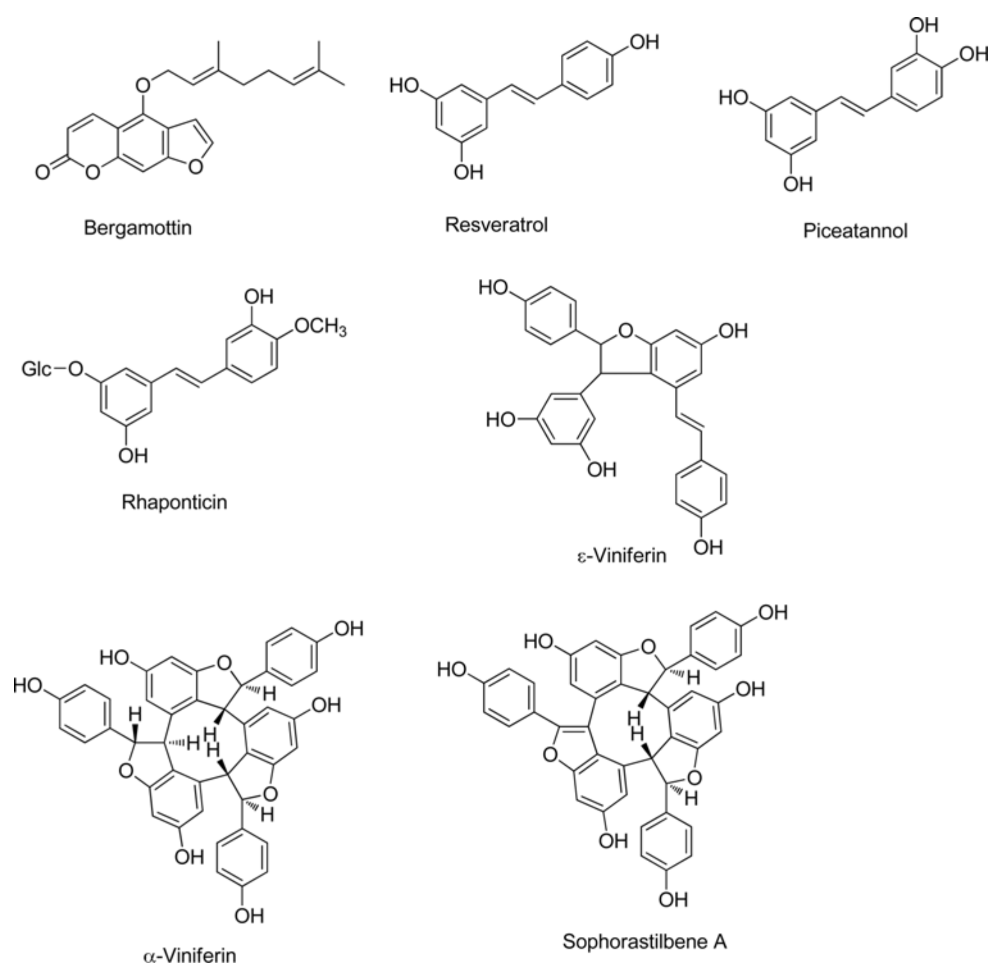

Fig. 3. The structures of the main active components of grape

Table III. Effects of grape extract and its active components on metabolic enzymes and transporters

Plant	*In vitro*/*In vivo*	Metabolism	Transporter	References
Grape				
Red wine solid	*In vitro* (H,R)	CYP2E1 (↓) CYP3A4 (↓) CYP3A (↓)		Piver et al., 2001
Resveratrol	*In vitro* (H, R)	CYP3A (↓) CYP3A4 (↓)		Piver et al., 2001
	In vivo (H)	CYP3A4 (↓) CYP2D6 (↓) CYP 2C9 (↓) CYP1A2 (↑) UGT1A1 (↔)		Chow et al., 2010
	In vitro (H)	UGT1A1 (↓) (s) UGT1A9 (↓) (s)		Brill et al., 2006
	In vivo (R)		MRP2 (s) BCRP (s)	Juan et al., 2010
	In vitro (H)		P-gp (↓)	Choi et al., 2009
ε-viniferin	*In vitro* (H)	CYP1A1 (↓) CYP1A2 (↓) CYP2B1 (↓) CYP2A6 (↓) CYP2B6 (↓) CYP3A4 (↓) CYP4A (↓)		Piver et al., 2003
bergamottin	*In vitro* (H)	CYP 1A2 (↓) CYP2A6 (↓) CYP2C9 (↓) CYP2C19 (↓) CYP2D6 (↓) CYP2E1 (↓) CYP3A4 (↓)		He et al., 1998

2006). Two main glucuronides, resveratrol-3-*O*-glucuronide and resveratrol-4'-*O*-glucuronide, were formed by UGT1A1 and UGT1A9, respectively, in human liver and intestinal microsomes. In addition, resveratrol inhibited glucuronidation via UGT1A1 and UGT1A9 when co-incubated with substrates for these enzymes. Hence, resveratrol may act as an inhibitor of UGT1A1 and 1A9 and influence glucuronidation of substrates derived from phytochemical components (Brill et al., 2006). Due to the low oral bioavailability of resveratrol, the effect of resveratrol on ABC transporters such as ABCB1, MRP2, and BCRP was assessed *in vivo* models. MRP2 and BCRP were detected as the 2 apical transporters involved in the efflux of resveratrol (Juan et al., 2010). Resveratrol significantly reduced rhodamine123 efflux via P-gp in MCF-7/ADR cells overexpressing P-gp, suggesting resveratrol as an inhibitor of P-gp (Choi et al., 2009). Resveratrol is metabolized into polar metabolites, resveratrol glucuronide and sulfate, in the gut epithelium. The transport of these metabolites was proposed to be related to BCRP (van de Wetering et al., 2009). A recent study disclosed that BCRP1 may play an important role in the efflux of resveratrol conjugates (Alfaras et al., 2010). Resveratrol oligomers, α-viniferin, sophorastilbene A, and ε-viniferin, may efficiently inhibit MRP1-mediated organic anion transport. This inhibitory potency of stilbenes seemed to increase with oligomerisation (Bobrowska-Hägerstrand et al., 2006). All chemical structures mentioned in this section, and the relevant metabolic enzymes and transporters are summarized in Fig. 3 and Table III, respectively.

Licorice

Licorice, the dried roots of *Glycyrrhiza glabra* or *G. uralensis*, refers, in China, Japan and Korea, to Glycyrrhizae Radix (Anonymous, 2011) and has a long history of medicinal use in Eastern and Western countries (Kinghorn and Compadre, 2001). Licorice extract has been used as an additive for flavoring and sweetening candies, chewing gums, toothpaste and beverages in Japan (Chin et al., 2007). Two main structural types, triterpenoids (e.g., glycyrrhizin) and flavonoids (e.g., liquiritigenin), are contained in the roots.

The administration of licorice extract or its main constituent, glycyrrhizin, daily, to mice for 1, 4, or 10 consecutive days affected CYP enzyme activities in different ways. In this experiment, a single dose did not modulate CYP enzymes while repeated treatment (4 or 10 days) of either extract or glycyrrizin was able to induce hepatic CYP3A4, 2B1, 1A2, 3A, 2A1, 2B1, and 2B9 (Paolini et al., 1998). Furthermore, the effect of glycyrrhizin on the activity of human CYP3A was determined in 16 volunteers and modest induction of

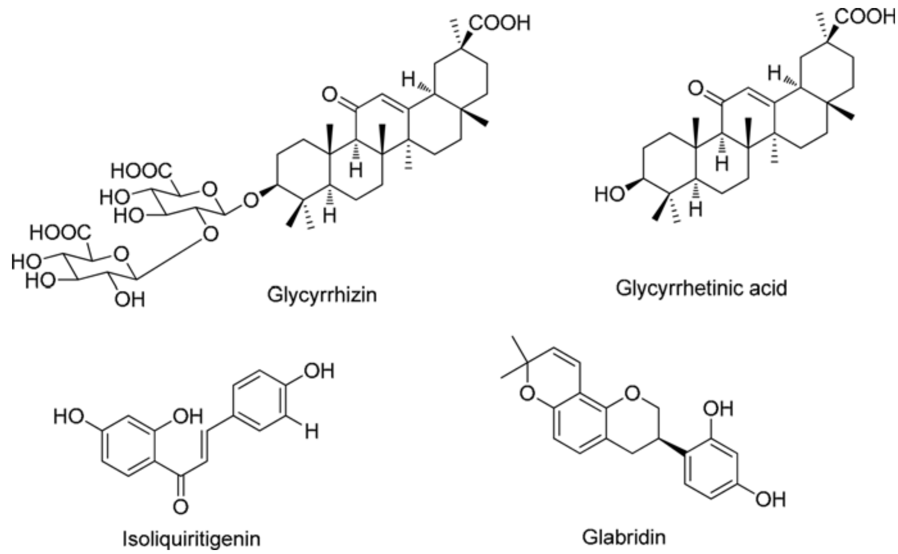

Fig. 4. The structures of the main active components of licorice

Table IV. Effects of licorice extract and its active components on metabolic enzymes and transporters

Plant	*In vitro*/*In vivo*	Metabolism	Transporter	References
Licorice				
Extract	*In vivo* (M)	CYP3A4 (↑) CYP2B1 (↑) CYP1A2 (↑) CYP3A (↑) CYP2A1 (↑) CYP2B1 (↑) CYP2B9 (↑)		Paolini et al., 1998
	In vivo (R)	UGT1A (↑)		Moon and Kim, 1997
Glycyrrhizin	*In vivo* (M)	CYP3A4 (↑) CYP2B1 (↑) CYP1A2 (↑) CYP3A (↑) CYP2A1 (↑) CYP2B1 (↑) CYP2B9 (↑)		Paolini et al., 1998
	In vivo (H)	CYP3A (↑)		Tu et al., 2010
	In vivo (R)	UGT1A (↑)		Moon and Kim, 1997
	In vitro (R)		P-gp (↓)	Najar et al., 2010
Glycyrrhetinic acid	*In vitro* (H)		P-gp (↓)	Nabekura et al., 2008
Glabridin	*In vitro* (H)	CYP3A4 (↓) CYP2B6 (↓) CYP2C9 (↓) CYP2D6 (↔) CYP2E1 (↔)		Kent et al., 2002
	In vitro (R)		P-gp (s) MRP1 (↓)	Yu et al., 2007
	In vitro (H)		P-gp (↔) MRP1 (↔)	Nabekura et al., 2008
Isoliquiritigenin	*In vivo* (R)	QR (↑) Glutathione (↑) Glutathione-S-transferase (↑)		Cuendet et al., 2010

CYP3A was observed (Tu et al., 2010). Besides glycyrrhizin, glabridin, an isoflavan-type compound, was investigated for its ability to modulate human CYPs (3A4, 2B6, 2C9, 2D6, 2E1) *in vitro* (Kent et al., 2002). Glabridin was found to inhibit human CYP 3A4, 2B6, and 2C9 activities. It was noted that inactivation of CYP3A4 was done by a loss of the intact heme moiety, but a loss of 2B6 enzyme activity was not related to the modification of the heme moiety. The enzymatic activities of CYP2D6 and 2E1 were not affected by treatment with glabridin. Isoliquiritigenin, a chalcone-type compound, induced quinone reductase-1 activity in the colon, and glutathione and glutathione-*S*-transferase in the liver (Cuendet et al., 2010). In the course of testing the effect of *G. glabra* extract on the metabolism of acetaminophen, it was proposed that this extract enhances glucuronidation. After administration of the extracts or glycyrrhizin to rats, activation of UGT1A was observed, but this activation did not result from the induction of UGT1A1 mRNA (Moon and Kim, 1997). P-gp was able to facilitate the efflux of glabridin in cultured RBMVECs, and when treated with verapamil, the level of glabridin was increased, suggesting P-gp as a transporter of glabridin (Yu et al., 2007). In addition to the involvement of P-gp as a transporter of licorice components, an inhibitor of P-gp and MRP1 was found in the licorice constituents. Glycyrrhetinic acid inhibited P-gp and MRP1 and reversed multidrug resistance while glabridin did not show any inhibitory effect (Nabekura et al., 2008). Glycyrrhizin, a sugar form of glycyrrhetinic acid, was also found to inhibit the ATPase activity of P-gp (Najar et al., 2010). All chemical structures mentioned in this section, and the relevant metabolic enzymes and transporters are summarized in Fig. 4 and Table IV, respectively.

Saw palmetto

Saw palmetto (*Serenoa repens*) preparations are one of the popular dietary supplements because of its effect on benign prostatic hyperplasia (Bent et al., 2006; Sinescu et al., 2011).

Saw palmetto extract was orally administered to healthy volunteers (12 persons) for 14 days and changes in CYP2D6 or 3A4 activity by this oral treatment was analyzed. Generally recommended doses were unlikely to alter the disposition of co-administered medications dependent on CYP2D6 or 3A4 (Markowitz et al., 2003b). Gurley et al. performed a clinical study regarding the activity of human CYPs for 28 days in 12 volunteers. There was no significant effect on CYP1A2, CYP2D6, CYP2E1, or CYP3A4 activity by pre-supplementation and post-supplementation (Gurley et al., 2004). Glucuronidation through UGT1A4 and UGT1A6 enzymes were apparently inhibited by saw palmetto extract in *in vitro* human liver microsomes (Mohamed and Frye, 2011b). All chemical structures mentioned in this section, and the relevant metabolic enzymes and transporters are summarized in Fig. 5 and Table V, respectively.

Garlic

For over five thousand years, garlic (*Allium sativum*) has been used as a therapeutic medicinal agent with a worldwide reputation. Garlic is reported to have antimicrobial, immune-enhancing, antiplatelet, hypolipi-

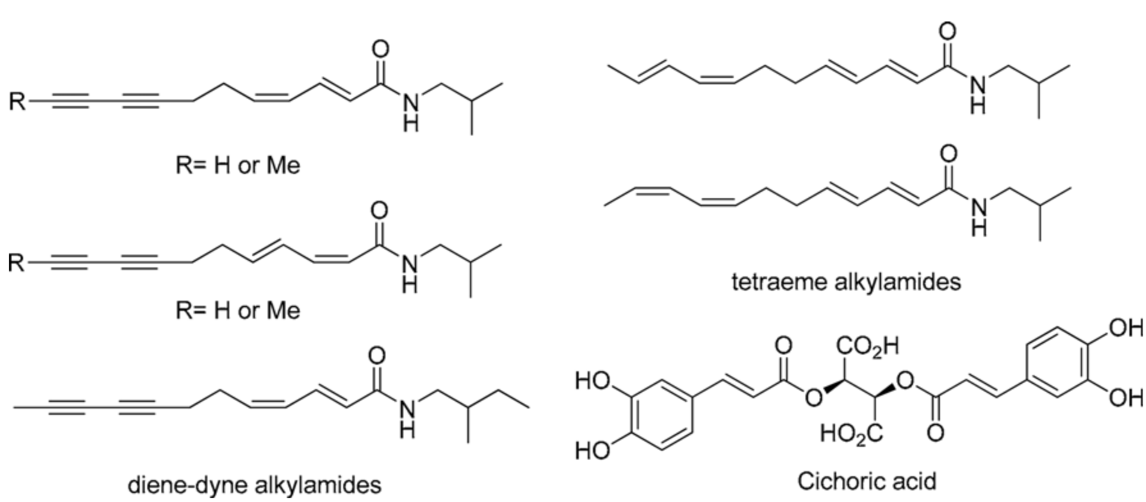

Fig. 5. The structures of the main active components of saw palmetto

Table V. Effects of saw palmetto extract and its active components on metabolic enzymes and transporters

Plant	*In vitro*/*In vivo*	Metabolism	Transporter	References
Saw palmetto				
Extract	*In vivo* (H)	CYP2D6 (↔) CYP3A4 (↔)		Markowitz et al., 2003b
	In vivo (H)	CYP1A2 (↔) CYP2D6 (↔) CYP2E1 (↔) CYP3A4 (↔)		Gurley et al., 2004
	In vitro (H)	UGT1A4 (↓) UGT1A6 (↓)		Mohamed and Frye, 2011b

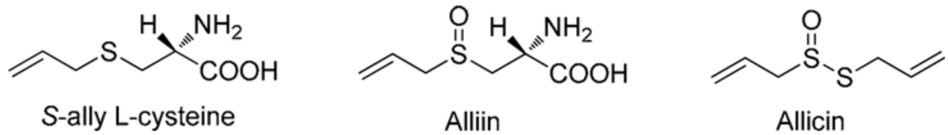

Fig. 6. The structures of the main active components of garlic

Table VI. Effects of garlic extract and its active components on metabolic enzymes and transporters

Plant	*In vitro*/*In vivo*	Metabolism	Transporter	References
Garlic				
Extract	*In vivo* (H)	CYP2D6 (↔) CYP3A4 (↔)		Markowitz et al., 2003c; Gurley et al., 2002; Kim et al.,1998
	In vitro (H)	CYP3A (↔) CYP2C9 (↔) CYP2C19 (↔) CYP3A4 (↔) CYP3A5 (↔) CYP3A7 (↔)		Foster et al., 2001
	In vitro (H)		P-gp (↑)	Kim et al., 1998; Manyike et al., 2000
Extract (w/o allicin)	*In vitro* (H)		P-gp (↔)	Dupuis et al., 2003
Garlic oil	*In vivo* (H)	CYP2E1 (↓)		Gurley et al., 2002; Gwilt et al., 1994
	In vivo (R)	CYP2C (↑)		Sparreboom et al., 2004
Garlic oil (w/o, allicin and alliin)	*In vivo* (M)	Glucuronidation of acetaminophen (↔)		Lin et al., 1996; Manyik et al., 2000
Diallyl sulfone	*In vivo* (M)	CYP2D6 (↔) CYP3A4 (↔)		Lin et al., 1996; Manyik et al., 2000
	In vivo (M)	CYP3A4 (↑) CYP2E1 (↑)	P-gp (↑)	Lee et al., 2006

demic and antitumor effects (Amagase et al., 2001; Harris et al., 2001; Kyo et al., 2001; Corzo-Martínez et al., 2007). It has been used to treat some opportunistic infections and is most commonly used by HIV-infected patients). Garlic contains a high-level of sulphur-containing compounds (e.g. allicin and alliin), numerous flavonoids/isoflavonoids (e.g. nobiletin, quercetin, rutoside and tangeretin), polysaccharides, prostaglandins, saponins and terpenes (e.g. citral, geraniol, linalool, and α- and β-phellandrene) (Dausch and Nixon, 1990; Singh et al., 2001). Organosulfur compounds from unstable thiosulfinates such as allicin, S-allylcysteine, S-allylmercaptocysteine and *N*-α-fructosyl arginine are also believed as active components for the beneficial effects of garlic (Amagase et al., 2001).

Garlic extract and/or single compound show various effects on CYP3A4 and/or P-gp systems. Garlic extract seemed to induce intestinal CYP3A4 and/or P-gp in clinical studies (Lee et al., 2006). For example, the bioavailability of saquinavir decreased by induction of P-gp in the gut mucosa with co-administration of garlic extract (Kim et al., 1998; Manyike et al., 2000). However, garlic extract (with allicin 600 μg, alliin 1.5 mg and *S*-allyl-L-cysteine 0.03 mg in each 600 mg tablet) did not affect the metabolism and excretion of dextromethorphan (a substrate of CYP2D6) (Markowitz et al., 2003c), alprazolam (a substrate of CYP3A4) (Markowitz et al., 2003c), midazolam (an intestinal CYP3A4 probe) (Gurley et al., 2002) and saquinavir (a substrate of CYP3A and P-gp) (Kim et al.,1998) in healthy volunteers. Similar results were observed for garlic oil without alliin and allicin; it did not alter the disposition of acetaminophen (primarily eliminated by CYP2D6- or CYP3A4-mediated metabolism (Manyik et al., 2000) and glucuronidation (Lin et al., 1996; Manyik et al., 2000) and reduced CYP2E1 activity (chlorzoxazone; a substrate for CYP2E1) (Gwilt et al., 1994; Gurley et al., 2002). Similarly, garlic extract without allicin had little effect on P-gp in human CD4 cells (Dupuis et al., 2003).

In rat studies, a single dose of garlic oil inhibited hepatic CYP2C and CYP2E1 activities, but administration of garlic oil for five days increased these hepatic CYP activities (Sparreboom et al., 2004). In human liver microsomes, garlic extract inhibited CYP3A activity, but did not affect CYP2C9, CYP2C19, CYP3A4, CYP3A5 and CYP3A7 activities (Foster et al., 2001). All chemical structures mentioned in this section, and the relevant metabolic enzymes and transporters are summarized in Fig. 6 and Table VI, respectively.

Milk thistle

Milk thistle (*Silybum marianum*) is one of the most commonly used herbal medicines for the treatment of

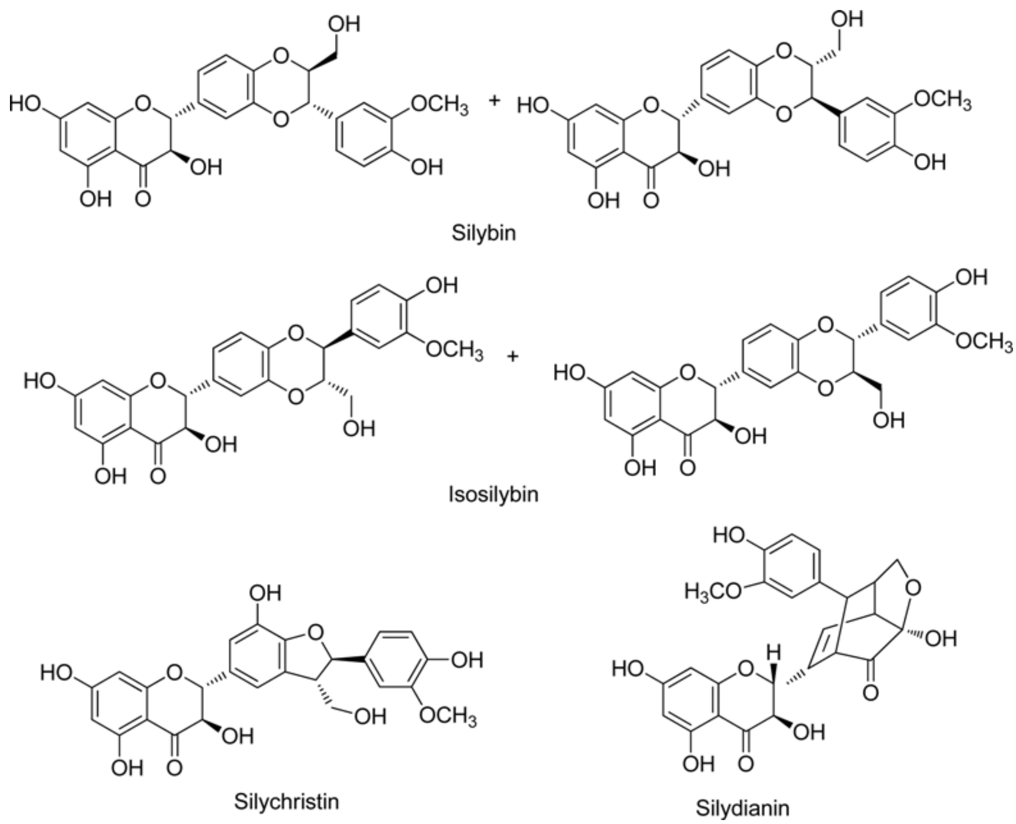

Fig. 7. The structures of the main active components of milk thistle

liver diseases (Ross, 2008). Milk thistle extract is rich in flavonolignans which involves silybin, silydianin and silychristine (Dhiman and Chawla, 2005). Silymarin is composed of mainly silibinin (silybin; 50-80%) with small amounts of other flavonolignans such as silychristin or silydianin (Quaglia et al., 1999; Ding et al., 2001; Kvasnicka et al., 2003). Milk thistle components have been shown to inhibit a variety of CYP isozymes, UGT and several efflux transporters (including ABCB1 and P-gp) (van Erp et al., 2005).

In vivo and *vitro* metabolism of milk thistle extract or of single compounds via CYP isozymes were determined using co-administered components. Milk thistle extract and silymarin showed negligible effects on CYP probes such as aminopyrine (a CYP nonspecific probe; 70 mg 3 times daily for 28 days) (Leber and Knauff, 1976), phenylbutazone (a CYP nonspecific probe; 70 mg 3 times daily for 28 days) (Leber and Knauff, 1976), indinavir (a CYP3A substrate; silymarin ingestion 160 mg 3 times daily for 20 days 153 mg) (Piscitelli et al., 2002; van Erp et al., 2005). These results suggest that silymarin and milk thistle extract did not inhibit CYP1A2, 2D6, 2E1, and 3A4 in *in vivo* human studies. However, *in vitro* investigations implicate milk thistle extract and/or silibinin as inhibitors of human CYP3A4 (Venkataramanan et al., 2000; Sridar et al., 2004), CYP2C9 (Zuber et al., 2002; Sridar et al., 2004), CYP2D6 (Zuber et al., 2002) and CYP2E1 (Zuber et al., 2002) (not CYP1A (Beckmann-Knopp et al., 2000)). Moreover, milk thistle intake does not affect systemic metabolism of UGT substrates because plasma concentrations of flavonolignans *in vivo* is not enough for the inhibition of drug glucuronidation (Sridar et al., 2004).

Silibinin is primarily conjugated (both glucuronidation and sulfation) and excreted in the bile and urine in rats (Lorenz et al., 1982) and humans (Schandalik et al., 1992; Weyhenmeyer et al., 1992; Gatti and Perucca, 1994; Kren et al., 2000). Thus, there would be competition between silibinin (involved in milk thistle effects) and endogenous/exogenous substances that are metabolised to glucuronides in the liver by UGTs. Thus, *in vitro* experiments showed inhibitory effects of milk thistle compounds on UGT enzymes; milk thistle inhibits glucuronidase (Kim et al., 1994), UBT systems such as UGT1A1, UGT1A6, UGT1A9, UGT2B7, and UGT2B15 isoforms (Sridar et al., 2004), ABCB1 (P-gp) (Zhang and Morris, 2003) and ABCC1 (MRP1) (Nguyen et al., 2003). However, *in vivo* evidence for enzyme-mediated or transporter-mediated interactions

Table VII. Effects of milk thistle extract and its active components on metabolic enzymes and transporters

Plant	*In vitro*/ *In vivo*	Metabolism	Transporter	References
Milk thistle				
Extract (Legalon)	*In vivo* (R)	CYP (↔)		Leber and Knauff, 1976
Extract	*In vitro* (H)	CYP3A4(↓) UGT1A6/9(↓)		Venkataramanan et al., 2000
Silymarin	*In vivo* (H)	CYP3A (↔) CYP1A2 (↔) CYP2D6 (↔) CYP2E1 (↔)		Piscitelli et al., 2002
	In vitro (H)	CYP1A2(↔) CYP2D6(↔) CYP2E1(↔) CYP3A4(↔)		Gurley et al., 2004
	In vitro (H)		P-gp (↓)	Zhang and Morris, 2003
	In vitro (H)		MRP1(↓)	Nguyen et al., 2003
	In vitro (H)	UGT1A1		Williams et al., 2002b
silybin	*In vitro* (H)	CYP3A4 (↓) CYP2C9 (↓) UGT1A1 (↓) UGT 1A6 (↓) UGT 1A9 (↓) UGT 2B7 (↓) UGT 2B15(↓)		Sridar et al., 2004
	In vitro (H)	CYP2E1(↔) CYP 2A6(↔) CYP 2B6(↔) CYP 2C19(↔) CYP 2D6(↔) CYP 3A4(↓) CYP 2C9(↓)		Jancová et al., 2007
Silybin, silymarin	*In vivo* (H)	β-glucuronidase(↓)		Kim et al., 1994
Silybin, silydaianin, silycristin, dehydrosilybin	*In vitro* (H)	CYP3A4 (↓) CYP2D6 (↓) CYP2E1 (↓)		Zuber et al., 2002
Silibinin	*In vitro* (H)	CYP3A4(↔) CYP2E1(↔) CYP2C19(↔) CYP1A2(↔) CYP2A6(↔) CYP2C9 (↓)		Beckmann-Knopp et al., 2000

caused by milk thistle constituents is less compelling (van Erp et al., 2005).

On the other hand, silybin showed *in vivo-in vitro* correlation; silybin is catalyzed by recombinant CYP2C8 (Jancová et al., 2007) and inhibits recombinant CYP2C9 in a mechanism-based manner (e.g. inhibition of CYP2C9-mediated *S*-warfarin 7-hydroxylation in human liver microsomes) (Beckmann-Knopp et al., 2000; Sridar et al., 2004). Silybin also inhibited CYP1A2 and 2C8 moderately, but showed little inhibition of CYP2E1, 2A6, 2B6, and 2D6 in human studies (Jancová et al., 2007). Therefore, concentrations of oral administered silybin may be sufficient to compete for CYP binding sites in the liver and gut wall (Gurley et al., 2004) and this apparent lack of *in vitro-in vivo* correlations was due to the following reasons: poor bioavailability, large inter-individual variations in silybin absorption, lower CYP binding affinities of silybin conjugates, interproduct variability in silymarin content, or poor dissolution characteristics of milk thistle dosage forms (Venkataramanan et al., 2000). Also, silybin inhibited *in vitro* recombinant UGT1A1, 1A6, 1A9, 2B7 and 2B15 and *in vivo* UGT systems (Sridar et al., 2004).

There are several reports considering drug-herb interactions via CYP, UGT and/or P-gp systems with milk thistle or its single compounds. With milk thistle extract, the decreased AUC of indinavir was due to the modulation of CYP3A and P-gp systems by milk thistle extract (Piscitelli et al., 2002; DiCenzo et al., 2003) and increased AUC of doxorubicin was due to the inhibition of P-gp-mediated cellular efflux by silymarin and its metabolites (Zhang and Morris, 2003). However, silibinin had little effect on the metabolism of erythromycin (CYP3A4), chlorzoxazone (CYP2E1), *S*(+)-mephenytoin (CYP2C19), caffeine (CYP1A2) or coumarin (CYP2A6) in human liver microsomes (Beckmann-Knopp et al., 2000), and did not significantly affect CYP1A2, CYP2D6, CYP2E1 or CYP3A4 activities in clinical studies (Gurley et al., 2004). All chemical structures mentioned in this section, and the relevant metabolic enzymes and transporters are summarized in Fig. 7 and Table VII, respectively.

Ginseng

Panax ginseng is one of the most sought-after medicines throughout the world, especially in Asia for thousands of years. Orally administered *Panax ginseng* has numerous pharmacological effects such as antihypertensive, antifatigue, neuroprotective, antioxidative, hypolipidemic, cognition-enhancing, immunoenhancing, ulcer-healing and anticancer effects (Xiang et al., 2008; Lü et al., 2009). Its major active constituents include ginsenosides (panoxosides, more than 150 isolated), sterols, flavonoids, peptides, vitamins, polyacetylenes, minerals, β-elemine, and choline (Christensen, 2009). Ginsenosides are considered major pharmacologically active constituents and approximately 12 types of ginsenosides have been isolated and structurally identified. The pharmacologically active components of ginseng are the final metabolites transformed by intestinal microflora. Compound K (20-*O*-β-D-glucopyranosyl-20*S*-protopanaxadiol) is a metabolite generated from ginsenosides Rb1, Rb2, Rc, and Rd; while ginsenosides Re, Rg1 and Rf are converted to protopanaxadiol (Christensen, 2009).

In human microsomal studies, *Panax ginseng* extract and individual ginsenosides showed various inhibitory and/or inducing effects on CYP. A standardized *Panax ginseng* extract inhibited the activity of CYP1A1/2 (Chang et al., 2002) and 1B1 (Chang et al., 2002) and 2E1 (concentration-dependent manner; Kim et al., 1997). On the other hand, the individual ginsenosides (Rb1, Rb2, Rc, Rd, Re, Rf, and Rg1) show the inhibitory effect on CYP1A1 only at a high concentration (over 100 μg/mL) (Chang et al., 2002). Ginsenoside Rd had only weak inhibitory activity against CYP3A4, CYP2D6, CYP2C19 and CYP2C9, whereas ginsenoside Re and ginsenoside Rf substantially increased the activity of CYP2C9 and CYP3A4 (Henderson et al., 1999). However, in clinical human studies, *Panax ginseng* extract is unlikely to alter the disposition of co-administered drugs that are primarily dependent on the CYP2D6 or CYP3A4 pathway (Donovan et al., 2003).

No reports of ginsenoside glucuronidation were found in the literature. In a pharmacokinetic study in which ginsenoside Rd was administered intravenously to volunteers, no glucuronidated metabolites were detected in the rat S9 liver fraction (Yang et al., 2007) and in clinical pharmacokinetic studies (Cai et al., 2003). All chemical structures mentioned in this section, and the relevant metabolic enzymes and transporters are summarized in Fig. 8 and Table VIII, respectively.

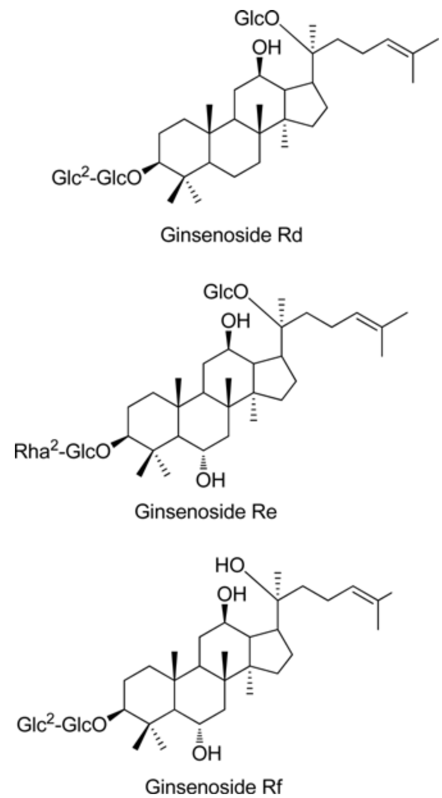

Fig. 8. The structures of the main active components of ginseng

St. John's wort

The use of St. John's wort has been increasing exponentially in the United States and Europe. St. John's wort is currently largely unregulated due to low rates of side effects as a dietary supplement. St. John's wort (*Hypericum* perforatum) extract is used for insomnia and depression (Gaster and Holroyd, 2000). A major class of compounds in St. John's wort extracts are flavonol glycosides with rutin, hyperoside, isoquercitrin, quercitrin (quercetin 3-rhamnoside) and miquelianin, hypericin, pseudohypericin and hyperforin (Butterweck and Schmidt, 2007). The mechanism for the antidepressant activity of St John's wort is unclear. Current evidence regarding the antidepressant effect of hypericum extracts is inconsistent and confusing (Kasper et al., 2008).

It was reported that St John's wort might inhibit or induce various CYPs and/or P-gp, depending on dose, route and duration of administration (Jiang et al., 2004). *In vitro* human microsomal studies have demonstrated that St John's wort inhibits CYP2C9, CYP2D6 and CYP3A4 activities (Obach, 2000; Wang et al., 2001). Paradoxically, in human studies St John's wort induced hepatic CYP1A2, hepatic CYP2E1 and hepatic/intestinal CYP3A4 (Roby et al., 2000; Wang et al., 2001; Johne er al., 2002; Mathijssen et al., 2002; Dresser et al., 2003) and MDR1/P-gp systems (Dürr et

Table VIII. Effects of ginseng extract and its active components on metabolic enzymes and transporters

Plant	*In vitro*/*In vivo*	Metabolism	Transporter	References
Ginseng				
Extract	*In vivo* (R)	CYP2E1 (↓)		Kim et al., 1997
	In vivo (H)	CYP2D6 (↔) CYP3A4 (↔)		
	In vitro (H)	CYP1A1 (↓) CYP1B1 (↓)		Chang et al., 2002
Ginsenoside Rb1, Rb2, Rc, Rd, Re, Rf	*In vitro* (H)	CYP1A1 (↓) CYP1A2 (↓) CYP1B1 (↓)		Chang et al., 2002
Ginsenoside Rd	*In vitro* (R)	CYP3A4(↓) CYP2D6(↓)		Henderson et al., 1999
		CYP2C19(↓) CYP2C9(↓)		Henderson et al., 1999
	In vitro (R)	Glucuronidation (s)		Yang et al., 2007
	In vivo (H)	Glucuronidation (s)		Cai et al., 2003
Ginsenoside Re, Rf	*In vitro* (R)	CYP2C9 (↑) CYP3A4 (↑)		Henderson et al., 1999

al., 2000). A similar result was reported in mice studies in which P-gp expression decreased after administration of St John's wort (Sugimoto et al., 2001).

With respect to the herb-drug interactions via CYP and/or P-gp systems, there were several case reports in human studies as followings. St John's wort decreased the blood concentrations of amitriptyline (due to the induction of CYP3A; Venkatakrishnan et al., 1999), cyclosporine (due to the induction of CYP3A and P-gp; Dresser et al., 2003), digoxin (due to the induction of P-gp (not CYP isozymes); Schinkel et al., 1995; Dresser et al., 2003), fexofenadine (due to the induction of P-gp (the opposite effect of a single dose; inhibition of intestinal P-gp); Dresser et al., 2003), indinavir (due to the induction of CYP3A; Piscitelli et al., 2000), methadone (due to the induction of CYP3A; Eich-Höchli et al., 2003), midazolam (due to the induction of CYP3A; Wang et al., 2001), simvastatin (due to the induction of CYP3A and P-gp; Sugimoto et al., 2001), tacrolimus (due to the induction of CYP3A and P-gp; Mai et al., 2003), theophylline (due to the induction of CYP1A; Nebel et al., 1999) and warfarin (due to the induction of CYP2C9; Jiang et al., 2004). On the other hands, it did not alter the pharmacokinetics of carbamazepine (Burstein et al., 2000), dextromethorphan (Markowitz et al., 2000) and pravastatin (Sugimoto et al., 2001).

Also the components of St John's wort are involved in glucuronidation such as quercetin and hypericin. Quercetin is a known substrate and modulator of UGT1A enzymes in human studies (Sun et al., 1998; Oliveira et al., 2000; van der Logt et al., 2003; Chen et al., 2008; Mohamed et al., 2010). Also hypericin competitively inhibited UGT1A6-mediated glucuronidation of acetaminophen in human colon cells and serotonin in UGT1A6-expressing insect cells, suggesting that the mechanism of this interaction was through

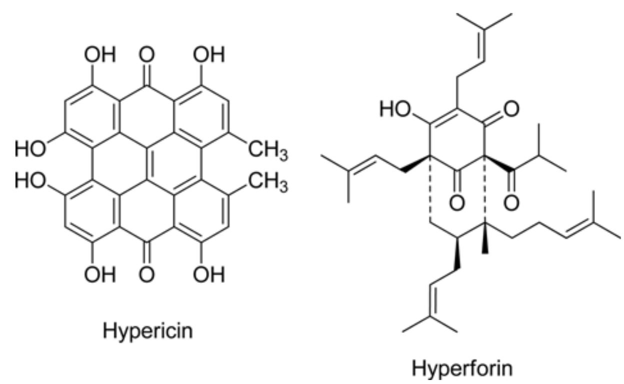

Fig. 9. The structures of the main active components of St. John's wort

the inhibition of UGT1A6 phosphorylation (Volak and Court, 2010). All chemical structures mentioned in this section, and the relevant metabolic enzymes and transporters are summarized in Fig. 9 and Table IX, respectively.

CONCLUSION

The aim of this review was to describe the effects of medicinal herbal products on drug metabolism and transporter enzymes. The effects of nine botanicals on metabolic and transporter enzymes are summarized in Table I-IV. The possible drug interactions by ginkgo extracts seem not as significant based on human *in vivo* studies, but most of the *in vitro* data showed inhibition of metabolism and transporter enzymes. Green tea extract also has inhibitory effects on CYPs and transporters *in vitro* studies, but human data is very limited. Red wine solids (grapes) and other active components including resveratrol inhibit the main phase I and II metabolic enzymes. Licorice extract and its main component, glycyrrhizin, induce most CYPs

Table IX. Effects of St John's wort extract and its active components on metabolic enzymes and transporters

Plant	*In vitro*/*In vivo*	Metabolism	Transporter	References
St John's wort				
	In vivo (H)	CYP1A2 (↑) CYP2E1 (↑) CYP3A4 (↑)		Dresser et al., 2003; Mathijssen et al., 2002; Johne er al., 2002; Roby et al., 2000; Wang et al., 2001
	In vitro (H)	CYP2C9 (↓) CYP2D6 (↓) CYP3A4 (↓)		Obach, 2000; Wang et al., 2001
	In vivo (H)	CYP3A (↑)		Venkatakrishnan et al., 1999
	In vivo (H)	CYP3A (↑)		Piscitelli et al., 2000
	In vivo (H)	CYP3A (↑)		Eich-Höchli et al., 2003
Extract	*In vivo* (H)	CYP3A (↑)		Wang et al., 2001
	In vivo (H)	CYP3A (↑)	P-gp (↑)	Dresser et al., 2003
	In vivo (H)	CYP3A (↑)	P-gp (↑)	Sugimoto et al., 2001
	In vivo (H)	CYP3A (↑)	P-gp (↑)	Mai et al., 2003
	In vivo (H)		MDR1/P-gp (↑)	Dürr et al., 2000
	In vitro/*vivo* (M)		P-gp (↓)	Sugimoto et al., 2001
	In vivo (H)		P-gp (↑)	Dresser et al., 2003; Schinkel et al., 1995
	In vivo (H)	CYP1A (↑)		Nebel et al., 1999
	In vivo (H)	CYP2C9 (↑)		Jiang et al., 2004
Hypericin	*In vitro* (H)	UGT1A6 (↓)		Volak and Court, 2010
Quercetin	*In vitro*/*In vivo* (H)	UGT1A (s, modulator)		Chen et al., 2008; Oliveira et al., 2000; Mohamed and Frye, 2010; Sun et al., 1998; van der Logt et al., 2003

of humans and rats, but glabridin shows inhibitory effects on some CYPs. Saw palmetto extracts do not affect most CYPs, but some phase II enzymes are inhibited by saw palmetto. Garlic and milk thistle extracts do not cause severe drug interactions in humans. There have been inconsistent result reported about sylibin, which is one of the active components of milk thistle. In the case of Ginseng, the extract and individual active component showed different effects on CYPs. Reports about St John's wort clearly suggest the need for careful use of this herb by patients who are taking other drugs that are metabolized or excreted by CYPs and P-gp.

There are a lot of challenges for the safe use of herbal medicines from natural products. First of all, efficacy and safety data should be produced for ensuring the safe use of the herbal medicine by pharmacologists, toxicologists and regulatory agencies. The exact quality-control (QC) for index components and possible active component in extract should be done for the clear comparison of herbal medicine. Patient-related factors such as pathophysiological status due to multiple diseases, co-pharmacotherapy, or genetic polymorphism should be carefully considered. A drug-drug interaction database already exists, and has been used for the purpose of the eliminating potential drug-drug interactions. The documentation of herbal medicinal products through a multi-pronged approach involving physicians, pharmacists and other healthcare practitioners could help optimize pharmacotherapy.

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